Assignment 3

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Q1.

- What is the class C address range in decimal and in Binary.?

→ Decimal range – 192.0.0.0 — 223.255.255.255
Binary range – 11000000 — 11011111

- What is the valid range of class A network address?

→ valid range – 1.0.0.1 — 126.255.255.254

- What is the class A private IP address space?

10.0.0.0 to 10.255.255.255

- What is the class B private IP address space

172.16.0.0 to 172.31.255.255

- What is the class C private IP address space.

192.168.0.0 to 192.168.255.255

Q2.

1) Which class of IP addresses uses the pattern shown here ?
(Network Network Network Host)

→ Class C

2) Which of following is an example of the multicast address?

→ C) 224.0.0.10

3) Which two of following are private Ir address.

C) 172.20.14.36

E) 192.168.24.43

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4. What is address range of class B network address in binary
C] 10xxxxxx

Q3.

Find out networks and broadcast address for IP.

① 192.168.10.0 /24
N/A - 192.168.10.0
B/A - 192.168.10.255

② 10.0.0.0/8
N/A - 10.0.0.0
B/A - 10.255.255.255

⑦ 172.172.150.150/16
N/A - 172.172.0.0
B/A - 172.172.255.255

⑧ 150.150.150.150/16
N/A - 150.150.0.0
B/A - 150.150.255.255

④ 200.200.200.200/24
N/A - 200.200.200.0
B/A - 200.200.200.255

⑤ 100.100.100.0/8
N/A - 100.0.0.0
B/A - 100.255.255.255

⑥ 100.100.255.6/8
N/A - 100.0.0.0
B/A - 100.255.255.255

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Q4. All N/w & host range subnet mask

① Ex. 2 - 192.168.10.0 in this we need 8 network.
$2^3 = 8$
additional network bit = 3
CIDR → 24+3 = /27 ← Total network bits
host bits = 32 - 27 = 5
Total host = $2^5 = 32$
subnet mask - 255.255.255.224
Net 1: 192.168.10.0 -
192.168.10.31
Net 2 = 192.168.10.32
192.168.10.63
Net 3 = 192.168.10.64
192.168.10.95

② Ex 3: 200.200.200.0 in this we need 10 networks.
$2^4 \rightarrow 16$
additional network bits = 4
CIDR → 24+4 = /28 ← total network bits
host bits = 32-28 = 4
Total hosts = $2^4 = 16$
subnet mask = 255.255.255.240
Net 1 = 200.200.200.0
200.200.200.95
Net 2: 200.200.200.96
200.200.200.31
Net 3: 200.200.200.32
200.200.200.47

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③ Ex 4: 220.200.200.0 in this we need 6 networks.

$2^3 = 8$

additional network bits = 3

CIDR = 24 + 3 = /27 ← total network bits.

Host bits = 32 - 27 = 5

Host = 2^5 = 32

subnet mask - 255.255.255.224.

Net 1: 220.200.200.0

220.200.200.31

Net 2: 220.200.200.32

A: 220.200.200.63

Net 3: 220.200.200.64

220.200.200.95

④ Ex 5: 192.168.10.0 in this we need 60 networks.

$2^6 = 64$

additional network bits = 6.

CIDR = 24 + 6 = /30 ← total network bits.

Host bits = 32 - 30 = 2

Hosts = $2^2 = 4$

subnet mask - 255.255.255.252.

Net 1: 192.168.10.0

192.168.10.3

Net 2: 192.168.10.4

192.168.10.7

Net 3: 192.168.10.8

192.168.10.12

- Network and host range subnet mask.

① Ex 6: 192.168.10.0/25

10000000

host bits = 7

$n = 2^1 = 2$

$h = 2^7 = 128$

subnet mask.

255.255.255.128

Net1 : 192.168.10.0

192.168.10.127

Net2 : 192.168.10.128

192.168.10.255

② Ex 7. 192.168.10.0/27

11100000

host bits = 5

$n = 2^3 = 8$

$h = 2^5 = 32$

Subnet Mask

255.255.255.224

Net 1 : 192.168.10.0

192.168.10.31

Net 2 : 192.168.10.32

192.168.10.63

Net 3 : 192.168.10.64

192.168.10.95

③ Ex: 8 – 192.168.10.0/31

11111110

host bits = 1

$n = 2^7 = 128$

$h = 2^1 = 2$

subnet mask

255.255.256.254

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Net 1. 192.168.10.0
192.168.10.1
Net 2. 192.168.10.2
192.168.10.3

④ Ex 1: 220.200.200.0/29

11111000

host bit = 3

$n = 2^5 = 32$

$h = 2^3 = 8$

Subnet Mask
255.255.255.248

Net 1: 220.200.200.0
220.200.200.7
Net 2: 220.200.200.8
220.200.200.15

Q5. finding n/w address from an IP address & subnet mask.

- 192.168.10.198 255.255.255.247
Invalid subnet mask.

Q6. Write subnet, Broadcast address, & a valid host range

① 192.168.100.25/30

n = 6 bits

h bits = 2

no. of host bits = $2^2 = 4$

192.168.100.0 — N/A

192.168.100.1 ┐ usable

192.168.100.2 ┘ range

192.168.100.3 — B/A

⋮

192.168.100.23 — N/A

192.168.100.24 ┐ valid host

195.168.100.25 ┘ range

195.168.100.26 — B/A

② 192.168.100.37/28

n = 4 bits

h = 4 bits

no. of host = $2^4 = 16$

192.168.100.0 — N/A

192.168.100.1 ┐ usable

192.168.100.14 ┘

192.168.100.15 — B/A

⋮

192.168.10.32 — N/A

192.168.10.33 ┐ valid

192.168.10.46 ┘ host range

192.168.10.47 — B/A

③ 192.168.100.66/27

n = 3

h = 5

no. of Hosts = $2^5 = 32$

192.168.100.0 — N/A

192.168.100.1

192.168.100.30

192.168.100.31 — B/A

⋮

192.168.100.64 — N/A

192.168.100.65 ┐ valid host

192.168.100.94 ┘ range

192.168.100.95 — B/A

④ 192.168.100.17/29

n = 5 bits

h = 3 bits.

no. of hosts = $2^3 = 8$

192.168.100.0 — N/A

192.168.100.1 ┐ Usable

192.16.100.6 ┘ range

192.168.100.7 — B/A

⋮

192.168.100.16 — N/A

192.168.100.17 ┐ valid host

192.168.100.22 ┘ range

192.168.100.23 — B/A

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Q6. Write subnet, Broadcast address and valid host range.

⑤ 192.168.100.99/26

n = 2

h = 6

no. of hosts = 2^6 = 64

192.168.100.0

192.168.100.63

⋮

192.168.100.64 — N/A

192.168.100.65 ┐ valid host
192.168.100.126 ┘ range

192.168.100.127 — B/A

⑥ 192.168.100.99/25

n = 1

h = ~~128~~ 7

no. of hosts = 128

192.168.100.0 — N/A.

192.168.100.1 ┐ valid host
192.168.100.126 ┘ range

192.168.100.127 — B/A

Q7. You have a class B network and need 29 subnets. What is your mask.

Default mask : 255.255.0.0 = 116

need 29 subnet

2^5 = 32

Binary : 11111111 11111111 11111000 00000000

Mask — 255.255.248.0

Q8. What is the Broadcast address of 192.168.192.10/29?

n = 5

h = 3

no. of hosts = $2^3 = 8$

192.168.192.8 — N/A

192.168.192.9 ┐
192.168.192.14 ┘ valid host range

192.168.192.15 — B/A for 192.168.192.10

Q9. How many hosts are available with a class C/29 Mask

n = 5 bits

h = 3 bits

total no of hosts = $2^3 = 8$

Reserved address : 2 (1 N/A & 1 B/A)

Usable Hosts are : 8 - 2 = 6

Q10. What is subnet for host ID 10.16.3.65/23?

$= 2^{8+7}$

$= 2^1 = 2$

Binary			
11111111	11111111	11111110	00000000

Subnet mask — 255.255.254.0

~~10.16.0.0~~ 10.0.0.0 Subnet

~~10.16.0.0~~

~~10.16.2.0~~ 10.0.2.255

~~10.16.~~ :

10.0.3.0 — N/A

10.0.3.1 ┐
10.0.4.254 ┘ Usable range

10.0.4.255 — B/A

Assignment 4.

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Q2. A company's IP address is 172.18.0.0/16. They need to create subnets for four sites with host requirements of 8000, 60, 900, 2000. Using VLSM, calculate the network address, subnet mask, usuable host range, and broadcast address for each site.

① $2^{13} = 8192$

host bit

network bits = 32 − 13 = 19

172.18.0.0 ┐
172.18.31.255 ┘ class

$8.8.2^3 2^5.8$

subnet mask: 255.255.255.224

N/A − 172.18.0.0

B/A − 172.18.31.255

usuable − 172.18.0.1 − 172.18.31.254

② for 2000

$2^{11} = 2048$

host bits = 11

net bits = 32 − 11 = 21

$8.8.2^5 2^3.8$

172.18.32.0 ┐
172.18.39.255 ┘ class

subnet mask − 255.255.255.248

N/A − 172.18.32.0

B/A − 172.18.39.255

Usuable − 172.18.32.1 to 172.18.39.254

③ for 900

$2^{10} = 1024$

host bits = 10

net bits = 32 − 10 = 22

$8.8.2^6 2^2.8$

172.18.40.0

172.18.43.255

Subnet mask:- 255.255.255.252

N/A − 172.18.40.0

B/A − 172.18.43.255

usuable − 172.18.40.1 to 172.18.43.254

④ for 60

$2^6 = 64$

host bits = 6

net bits = 32 − 6 = 26

8 · 8 · 8 · 2^2 2^6

172.18.44.0

172.18.44.63

Subnet mask: 255.255.255.192

N/A − 172.18.44.0

B/A − 172.18.44.63

usable − 172.18.44.1 to 172.18.44.62.

Q1. Your company's IP address is 192.168.10.0/24. You need to create subnets for four departments with host requirements of 120, 80, 60, 14, and 6. Calculate the network address, subnet mask, usable host range, and broadcast address for each department.

192.168.10.0/24

① for 120 departments.

2^7 = 128

host bits

network bits = 32 − 7 = 25

8 · 8 · 8 · 2^1 2^7

class [192.168.10.0 / 192.168.10.127

subnet mask

255.255.255.128

N/A − 192.168.10.0

B/A − 192.168.10.127

usable [192.168.10.1 to 192.168.10.126

② for 360 departments.

2^6 − 64

host bits

network bits = 32 − 6 = 26

8 · 8 · 8 · 2^2 2^6

class [192.168.10.128 / 192.168.10.191

subnet mask − 255.255.255.192

N/A − 192.168.10.128

B/A − 192.168.10.191

Usable [192.168.10.129 to 192.168.10.190

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③ for 30 Departments.

$2^5 = 32$
host bits

network bits = 32 − 5 = 27

$8 \cdot 8 \cdot 8 \cdot 2^3 \; 2^5$

class [192.168.10.192 . 192.168.10.223]

Subnet mask − 255.255.255.224

N/A − 192.168.10.192

B/A − 192.168.10.223

Usable [192.168.10.193 , 192.168.10.222]

④ for 14

$2^4 = 16$

host bit = 4

network bits = 32 − 4 = 28

$8 \cdot 8 \cdot 8 \cdot 2^4 \; 2^4$

class [192.168.10.224 , 192.168.10.239]

subnet mask − 255.255.255.240

N/A − 192.168.10.224

B/A − 192.168.10.239

Usable [192.168.10.225 , 192.168.10.238]

⑤ for 6.

$2^3 = 8$

Host bits = 3

network bits = 32 − 3 = 29

$8 \cdot 8 \cdot 8 \cdot 2^5 \; 2^3$

class [192.168.10.240 , 192.168.10.247]

Subnet mask − 255.255.255.248

N/A − 192.168.10.240

B/A − 192.168.10.247

Usable [192.168.10.241 , 192.168.10.246]

Q3: Your Organization uses the IP address 10.0.0.0/8. You need to create subnets for each three branch offices with 50000, 20000, 5000 hosts. Solve this VLSM problem and provide the subnet details for each office.

10.0.0.0/8

① for 50000 hosts

$2^{16} = 65536$

host bits = 16

network bits = 32 - 16 = 16

8.8 2^8 2^0 .8.8.

class [10.0.0.0
10.0.255.255

subnet mask: ~~255.0.255.255~~
255.255.0.0

N/A — 10.0.0.0

B/A — 10.0.255.255

usable [10.0.0.1
10.0.255.254

② for 20000 hosts.

$2^{15} = 32766$

host bits = 15

network bits = 32 - 15 = 17

8.2^1 2^7 .8.8.

class [10.0.255.255
10.1.127.255

subnet mask — 255.255.128.0

N/A — 10.0.255.255

B/A — 10.1.127.255

usable [10.1.0.0
10.1.127.254

③ for 5000 hosts.

$2^{13} = 8192$, host bit = 13

net bits = 32 - 13 = 19 →

8.2^4 2^4 .8.8

class [10.1.128.0
10.1.128.